E commerce Product Entity Resolution Platform

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DECLARATION

The Mini Project Report entitled "**Project Title"** is a record of Bonafide work of N.yegnasaireddy-2520080055 ,**suresh babu-2520090036** submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering/Computer Science and Information Technology/ Artifical Intelligence & data Science to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is to certify that the mini project based report entitled "**Project Title"** is a bonafide work done and submitted by **N.yegnasairedddy-2520080055,Suresh babu-2520090036** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering/ Computer Science and Information Technology/Artificial Intelligence & Data Science, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

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1.ABSTRACT

The E-Commerce Product Entity Resolution & Matching Platform is a web-based project that applies Data Structures and Algorithms (DSA) to identify identical products listed by different online sellers. In e-commerce marketplaces, the same product may appear under different names, descriptions, specifications and prices, making product comparison difficult. The proposed system addresses this problem by normalizing product names and comparing important attributes such as brand, model, storage capacity and color. It uses Map-based hashing for product grouping, Set-based duplicate detection, arrays for storing product listings, string matching for comparing product names, and searching and sorting algorithms for efficient retrieval and price comparison. The frontend is developed using React.js, while Node.js and Express.js handle backend processing and REST API communication. When users search for a product, the platform displays the matching product group, available seller listings and the lowest price in the sample dataset. The project demonstrates the practical application of fundamental DSA concepts in solving an e-commerce data-matching problem. It also provides a foundation for future improvements such as database integration, advanced fuzzy matching and real-time price updates.

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2. INTRODUCTION

E-commerce platforms allow customers to purchase products from multiple sellers. However, the same product may be listed under different names, descriptions and formats. For example, “Apple iPhone 15 128GB Black” and “iPhone 15 Black 128 GB” may refer to the same product even though their text formats differ. These variations can create duplicate listings and make it difficult for customers to compare prices accurately.

Product entity resolution is the process of determining whether different records represent the same real-world product. This project proposes a web-based solution that normalizes product information, compares important attributes and groups matching product listings. It also provides search functionality and arranges seller listings by price to identify the lowest price in the available dataset.

The project uses React.js for the frontend and Node.js with Express.js for the backend. Its core DSA implementation uses Map-based hashing, Set-based duplicate detection, arrays, string matching, searching and sorting. The main objective is to demonstrate how these techniques can organize product information and support efficient product comparison. The following sections describe the related literature, project requirements, implementation modules, output screens, testing, conclusions and future enhancements.

3. LITERATURE SURVEY

1. Data Matching and Duplicate Detection — Peter Christen (2012)

This book explains data matching, record linkage, entity resolution and duplicate detection. It describes stages such as data preprocessing, indexing, field comparison, classification and evaluation. These concepts are directly relevant to identifying product records that refer to the same item. Our project adopts a simplified version of this workflow using text normalization, product attributes and DSA-based grouping.

2. Frameworks for Entity Matching: A Comparison — Köpcke and Rahm (2010)

This research paper compares multiple entity-matching frameworks and examines how different matching methods can be combined. It discusses candidate selection, matching strategies and evaluation approaches. The study highlights the importance of balancing matching effectiveness

and efficiency. Our project applies a simpler matching strategy suitable for a student-level DSA implementation.

3. Data Matching and Record Linkage — Australian National University Resource

This research resource explains how records from different sources can be linked when they represent the same entity. It discusses practical challenges in matching records and improving scalability. These ideas are relevant when comparing product listings collected from different sellers. Our project uses normalization and attribute comparison to organize similar product records.

4. React.js — Official Documentation

React.js documentation explains component-based interface development and the use of state to update displayed information. These features are useful for building product cards, search controls and seller comparison sections. Our proposed frontend uses React components to present matched products and price information. React provides the interface, while the backend handles product-processing operations.

5. Node.js and Express.js — Official Documentation

Node.js supports server-side JavaScript execution, while Express.js provides tools for creating web routes and APIs. These technologies can connect the frontend to the product-matching logic. In our project, API endpoints handle product retrieval and searching, and the DSA modules process product records before returning results to the interface.

4. Problem Statement and Objectives

Problem Statement: To develop an efficient E-Commerce Product Entity Resolution and Matching Platform using Data Structures and Algorithms to identify duplicate product listings, group identical products from different sellers, and compare their prices.

Problem Description: E-commerce platforms often contain multiple listings of the same product with variations in product names, descriptions, specifications, and prices. These variations make it difficult to identify identical products and compare prices accurately. The proposed system addresses this problem by normalizing product information, comparing product attributes, grouping matching listings, and identifying the lowest available seller price. Data Structures and

Algorithms are used to improve the organization, searching, matching, and sorting of product records.

Objectives:

1. To develop a web-based platform for e-commerce product entity resolution and matching.
2. To normalize product names and specifications to reduce differences in text formats.
3. To identify and group similar product listings from multiple sellers.
4. To detect duplicate product records using Set-based duplicate detection.
5. To use Map-based hashing for efficient product grouping and retrieval.
6. To apply string-matching techniques to compare product names and attributes.
7. To implement product searching and price sorting to identify the lowest available price.
8. To display matched product listings, matching scores, seller information, and price-comparison results through a user-friendly interface.
9. To demonstrate the practical application of Data Structures and Algorithms in an e-commerce system.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- Processor: Intel Core i3 or equivalent.
- RAM: Minimum 4 GB; 8 GB recommended.
- Storage: At least 500 MB of free space.
- Display: Standard laptop or desktop display.
- Keyboard and mouse.
- Internet connection for installing packages and deploying the project.

Software Requirements:

- Operating System: Windows 10/11, Linux or macOS.

- Frontend: React.js, JSX and CSS.
- Backend: Node.js and Express.js.
- Programming Language: JavaScript.
- Development Environment: Visual Studio Code.
- Package Manager: npm.
- Browser: Google Chrome, Microsoft Edge or Mozilla Firefox.
- Version Control: Git and GitHub (optional).

6. IMPLEMENTATION

The E-Commerce Product Entity Resolution & Matching Platform is organized into frontend, backend and DSA processing modules. The React.js frontend provides the product search interface and displays the matched product groups. The Node.js and Express.js backend receives requests and returns product information through REST APIs.

Module 1: Product Data Management

This module stores product information such as product name, brand, specifications, seller and price. A sample dataset is used to demonstrate how the same product can be listed by multiple sellers.

Module 2: Product Normalization and Matching

This module converts product names into a consistent format by changing text to lowercase and removing unnecessary spaces and symbols. The normalized information is then compared to identify potential product matches.

Module 3: Duplicate Detection and Product Grouping

This module uses Map-based grouping to organize related product listings and Set-based duplicate detection to prevent repeated records. Arrays store the seller listings associated with each matched product group.

Module 4: Product Search and Price Sorting

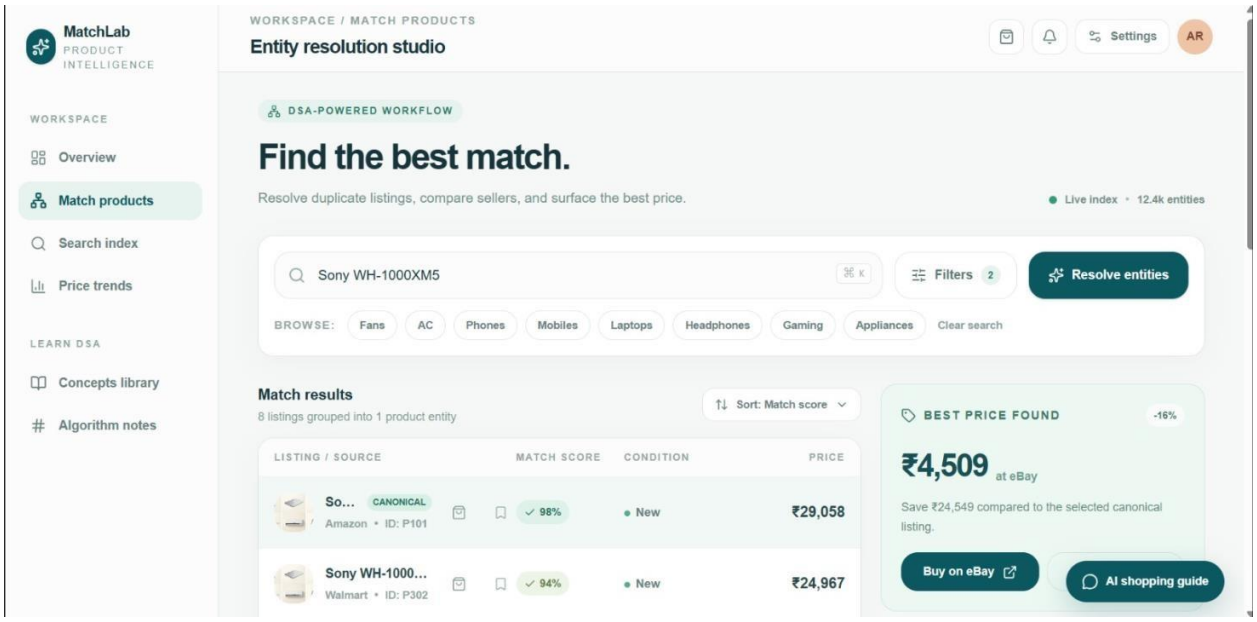
This module searches product records based on the user's query and sorts the seller listings by price. The lowest price in the available dataset is highlighted to help users compare the matched listings.

Implementation Flow:

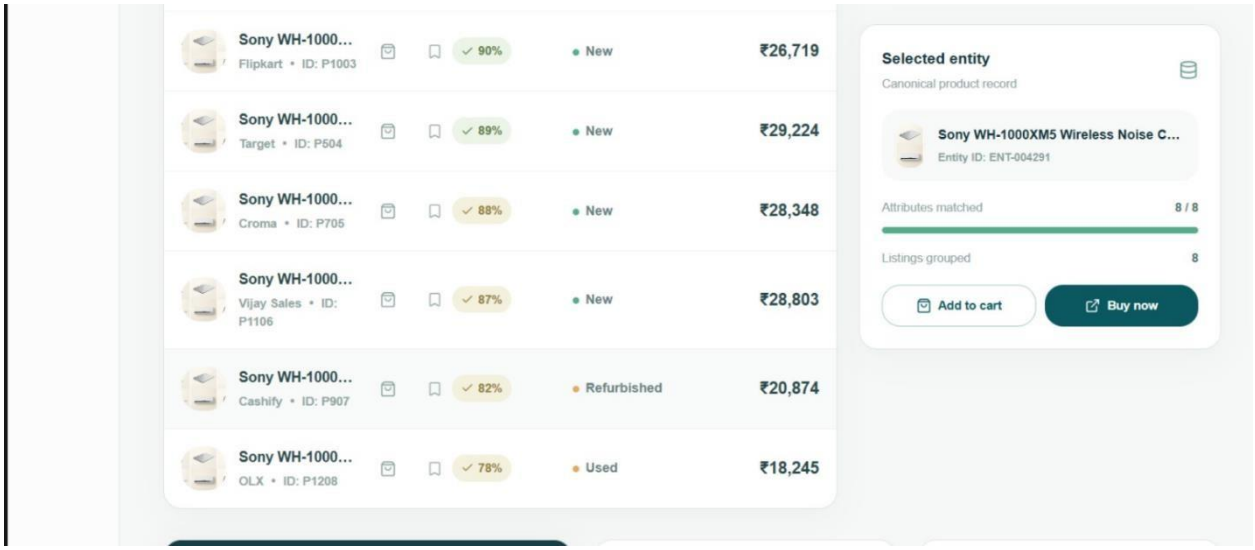
Product Dataset → Normalization → Product Matching → Duplicate Detection → Product Grouping → Searching and Sorting → React Website Output.

7. RESULTS & DISCUSSION

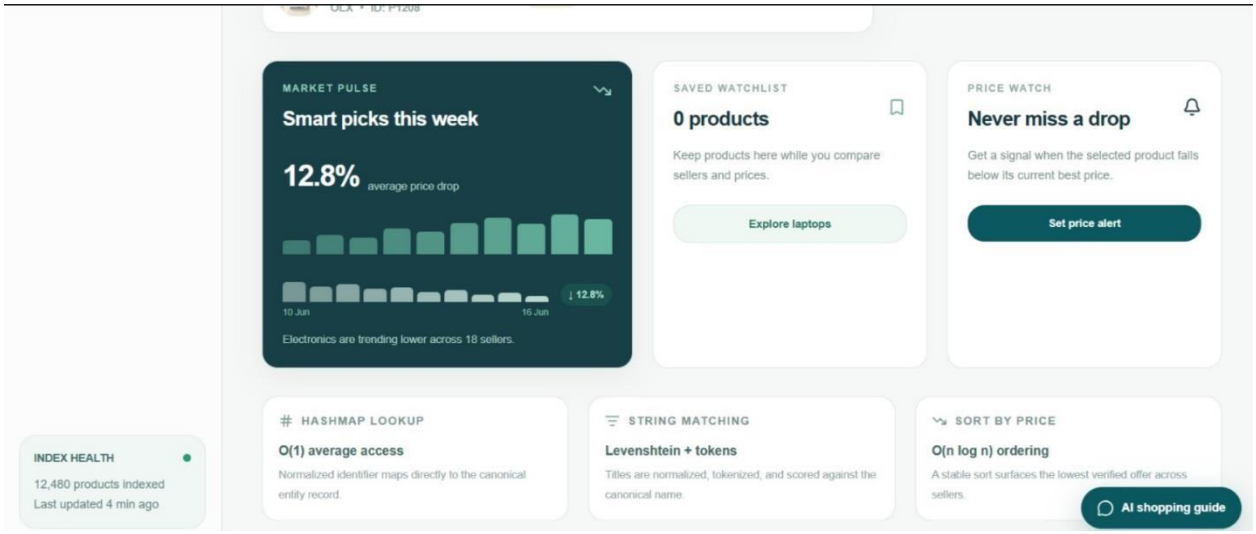
The application provides a product entity resolution interface that groups similar product listings, compares seller prices and presents information related to Data Structures and Algorithms. The following screenshots illustrate the matching results, selected product entities and DSA-related dashboard features.



The screenshot shows the product-matching interface, where users can search for products and resolve duplicate listings. It also displays match results, matching scores, seller prices and the best-price summary.



The screenshot displays multiple seller listings grouped under a selected Sony WH1000XM5 product entity. Match scores, product conditions and prices are shown to help compare listings and identify potential duplicates.



The dashboard presents price-trend information, watchlist and price-alert features, alongside explanations of HashMap lookup, string matching and price sorting. These panels illustrate how DSA concepts relate to product matching and price comparison.

8. CONCLUSION & FUTURE SCOPE

Conclusion

The E-Commerce Product Entity Resolution & Matching Platform demonstrates how Data Structures and Algorithms can be applied to a practical e-commerce problem. The project focuses on normalizing product information, identifying matching product listings, detecting duplicates and comparing seller prices. Map-based grouping, Set-based duplicate detection, arrays, string matching, searching and sorting form the core of the solution. React.js provides the user interface, while Node.js and Express.js support backend processing and API communication. The project illustrates how these techniques can organize product records and help users identify the lowest price within a given dataset. The effectiveness of the system depends on the quality of the matching logic and the accuracy of the available product information.

Future Scope

1. Database Integration: Integrate MySQL or MongoDB to store product information permanently.
2. Advanced Matching: Implement edit distance or other fuzzy string-matching methods to handle spelling variations and incomplete product names.
3. Real-Time Prices: Connect authorized product data APIs to retrieve updated seller prices.
4. Product Management: Add administrator functionality to add, update and remove product listings.
5. Improved Interface: Add product images, category filters and additional sorting options.
6. Performance Optimization: Evaluate the time and space complexity of matching and searching algorithms on larger datasets.
7. Deployment: Host the React frontend and Node.js backend online so users can access the application through a public URL.

9. REFERENCES

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